Subbases

Definition: Let $(X, \mathcal{T})$ be a topological space. A subclass $\mathcal{S}$ of $\mathcal{T}$ is called a subbase for $\mathcal{T}$ iff finite intersections of members of $\mathcal{S}$ form a base for $\mathcal{T}$, i.e. iff

$$\mathcal{B} = \mathcal{S}^{\cap} = \{\cap \mathcal{S}' \mid \mathcal{S}' \subseteq \mathcal{S} \ \& \ \mathcal{S}' \text{ is finite}\}$$

is a base for $\mathcal{T} \iff$ For every $U \in \mathcal{T}$ & every $x \in U$ there exists a $B = \cap \mathcal{S}' \in \mathcal{B}$, where $\mathcal{S}' \subseteq \mathcal{S}$ is finite, s.t. $x \in B \subseteq U$, i.e. there exist $S_1, S_2, \ldots, S_n \in \mathcal{S}$ s.t. $x \in S_1 \cap S_2 \cap \ldots \cap S_n \subseteq U$.

Theorem: Let (X, τ) & (X, τ') be topological spaces & $\mathcal{S}$ be a subbase for τ. If $\mathcal{S} \subseteq \tau'$ then $\tau \subseteq \tau'$. In other words the topology generated by a subbase $\mathcal{S}$ is the coarsest topology on X containing $\mathcal{S}$.

Proof: Assume that $\mathcal{S} \subseteq \tau'$.

Since τ' is a topology on X, it possesses (T2) & so $\bigcap \mathcal{S}' \in \tau'$ for any finite subclass $\mathcal{S}'$ of $\mathcal{S}$.

Hence $\mathcal{S}^\cap \subseteq \tau'$.

Similarly τ' possesses (T3) & so $\bigcup \mathcal{B}' \in \tau'$ for every subclass $\mathcal{B}'$ of $\mathcal{S}^\cap$.

But

$$\tau = \{ \bigcup \mathcal{B}' \mid \mathcal{B}' \subseteq \mathcal{S}^\cap \}.$$

$\therefore \tau \subseteq \tau'$.

Example (*) If $\mathcal{S}$ is a subbase for two topologies τ & τ' on X then $\tau = \tau'$.

Example: Let $(\mathbb{R}, \tau)$ be the usual topological space & $\mathcal{S} = \{(-\infty, b) \mid b \in \mathbb{R}\} \cup \{(a, \infty) \mid a \in \mathbb{R}\}$. Show that $\mathcal{S}$ is a subbase for τ.

Solution: Let τ' be the topology on $\mathbb{R}$ generated by $\mathcal{S}$.

Recall that $\mathcal{B} = \{(a,b) \mid a, b \in \mathbb{R}, a < b\}$ is a base for τ.

We shall show that $\tau = \tau'$.

- Let $a, b \in \mathbb{R}$ with $a < b$. Since

$(a,b) = (-\infty, b) \cap (a, \infty)$, $(a,b) \in \tau'$ since $\mathcal{S} \subseteq \tau'$ & τ' satisfies (T2)

Because τ' satisfies (T3) and $\mathcal{B}$ is a base for τ

any τ-open set U is also τ'-open.

$\therefore \tau \subseteq \tau'$ (1)

- Let $a, b \in \mathbb{R}$.

$$(-\infty, b) = \bigcup_{n=0}^{\infty} (b-n-2, b-n)$$

$$(a, \infty) = \bigcup_{n=0}^{\infty} (a+n, a+n+2)$$

i.e. $(-\infty, b)$ & (a, ∞) can be expressed as unions of base elements of τ & so they are τ-open, i.e. $\mathcal{S} \subseteq \tau$.

$(\mathbb{R}, \tau)$, $(\mathbb{R}, \tau')$ are topological spaces, $\mathcal{S}$ is a subbase for τ' & $\mathcal{S} \subseteq \tau$. Hence

$\tau' \subseteq \tau$ (2)

because τ' is the coarsest topology on $\mathbb{R}$ as it is generated by a subbase $\mathcal{S}$.

$\therefore$ By (1) & (2) $\tau = \tau'$. $\mathcal{S}$ is a subbase for τ.

Let $\{A_i\}_{i\in I}$ be a collection of sets. Define intersection of $\{A_i\}_{i\in I}$ as

$$\cap_{i\in I} A_i = \{x \mid \text{for every } \ i \in I, x \in A_i\}.$$

When $I = \emptyset$ this definition becomes

$$\cap_{i\in \emptyset} A_i = \{x \mid \text{for every } \ i \in \emptyset, x \in A_i\}.$$

Let X be the universal set and $x \in X$. The statement "for every $i \in \emptyset, x \in A_i$" is vacuously true since there are no indices i in $\emptyset$ that can make $x \in A_i$ false. Therefore if $\mathcal{A} = \{A_i \mid i \in \emptyset\}$ then

$$\cap_{i\in \emptyset} A_i = \cap \mathcal{A} = X.$$

Sol^n It follows from the previous theorem.

$\longleftrightarrow$

Let $\mathcal{S}$ be a class of subsets of a nonempty set X. $\mathcal{S}$ may or may not be a base for a topology on X. However $\mathcal{S}$ always generates a topology on X in the following sense:

Theorem: Any class $\mathcal{S}$ of subsets of a nonempty set X is the subbase for a unique topology τ on X. That is finite intersections of members of $\mathcal{S}$ form a base for the topology τ on X.

Proof: Let $\mathcal{B} = \mathcal{S}^n$.

We wish to prove that $\mathcal{B}$ is a base for a topology on X. But then it suffices to show that $\mathcal{B}$ possesses (B1) & (B2).

(B1) Since the empty class

$\mathcal{A} = \{A_i \mid i \in \emptyset\}$ is a finite subclass of $\mathcal{S}$ & $\bigcap \mathcal{A} = X$, $X \in \mathcal{B} = \mathcal{S}^\cap$.

Hence $\bigcup \mathcal{B} = X$.

(B2) Let $B_1, B_2 \in \mathcal{B}$ & $x \in B_1 \cap B_2$.

Then there exist finite subclasses $\mathcal{S}_1$ & $\mathcal{S}_2$ of $\mathcal{S}$ s.t. $B_1 = \bigcap \mathcal{S}_1$ & $B_2 = \bigcap \mathcal{S}_2$. Let $\mathcal{S}_3 = \mathcal{S}_1 \cup \mathcal{S}_2$.

Then $\mathcal{S}_3$ is a finite subclass of $\mathcal{S}$ & hence $B_3 = \bigcap \mathcal{S}_3 \in \mathcal{B} = \mathcal{S}^\cap$.

Moreover $x \in B_3 \subseteq B_1 \cap B_2$.

($B_1 = \bigcap \mathcal{S}_1$ & $B_2 = \bigcap \mathcal{S}_2 = S_1' \cap S_2' \cap \dots \cap S_m'$

$= S_1 \cap S_2 \cap \dots \cap S_n$

$x \in B_1 \cap B_2 = S_1 \cap S_2 \cap \dots \cap S_n \cap S_1' \cap S_2' \cap \dots \cap S_m'$

$\mathcal{S}_3 = \mathcal{S}_1 \cup \mathcal{S}_2 = \{S_1, \dots, S_n, S_1', S_2', \dots, S_m'\}$.

$B_3 = \bigcap \mathcal{S}_3 = S_1 \cap \ldots \cap S_n \cap S_1' \cap \ldots \cap S_n'$

$x \in B_3 \subseteq B_1 \cap B_2$)

The uniqueness of τ follows from Example (*) given above.

Example: Let $X = \{1, 2, 3, 4\}$ &

$\mathcal{S} = \{\{1\}, \{3\}, \{2, 4\}, \{1, 3, 4\}\}$. Find the topology τ on X generated by $\mathcal{S}$.

Sol^n

$\mathcal{B} = \mathcal{S}^{\cap} = \{\phi, \{1\}, \{3\}, \{2, 4\}, \{4\}, \{1, 3, 4\}, X\}$ is a base for a topology τ on X.

Let us find τ.

$\tau = \{\phi, \{1\}, \{1, 3\}, \{1, 2, 4\}, \{1, 4\}, \{1, 3, 4\}, X,$
$\{3\}, \{2, 3, 4\}, \{3, 4\}, \{2, 4\}, \{4\}\}$